

UAV ASSIGNMENT 01

NILAVRA GHOSH

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Contents

1	Q1	3
1.1	Concept of Operations (CONOPS)	3
1.2	Requirement Specifications	4
1.2.1	Primary Mission Requirements	4
1.2.2	Derived Performance Requirements	4
1.2.3	Payload Force Requirement	5
1.2.4	Operational and Safety Requirements	5
1.3	Market Survey	6
1.3.1	Surveyed UAV 1: EHang EH216-S	6
1.3.2	Surveyed UAV 2: Volocopter 2X	6
1.4	Sizing and Layout	8
1.4.1	Initial Weight Estimation	8
1.4.2	Rotor Thrust Distribution	8
1.4.3	Rotor and Frame Dimensioning	8
1.4.4	Disk Loading Calculation	9
1.5	Component Identification	10
1.5.1	Airframe Structure	10
1.5.2	Rotor System	10
1.5.3	Electric Propulsion System	10
1.5.4	Energy Storage System	11
1.5.5	Flight Control System	11
1.5.6	Navigation and Communication System	11
1.5.7	Payload System	11
1.5.8	Landing Gear	11
1.6	Performance Analysis	11
1.6.1	Hover Performance	11
1.6.2	Hover Power Requirement	12
1.6.3	Climb Performance	12
1.6.4	Cruise Power Requirement	13
1.6.5	Energy Requirement for Mission	13
1.7	Optimization for Maximum Endurance	13
1.7.1	Reduction of Disk Loading	14
1.7.2	Weight Optimization	14
1.7.3	Propulsion Efficiency Improvement	14
1.7.4	Cruise Power Optimization	14
1.7.5	Rotor Configuration Selection	14
1.7.6	Optimized Design Outcome	15
2	Q2	15
2.1	Propeller 1: APC 9×6	19
2.1.1	Case 1: 4000 RPM	19
2.1.2	Case 2: 5000 RPM	20
2.1.3	Case 3: 6000 RPM	21
2.2	Propeller 2: APC 10×7	22

2.2.1	Case 1: 4000 RPM	22
2.2.2	Case 2: 5000 RPM	23
2.2.3	Case 3: 6000 RPM	24
2.2.4	Comparison of Propeller Performance	24
2.2.5	Discussion	25
2.2.6	Conclusion	26



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H Y D E R A B A D



1 Q1

So, we are given a few design requirements and am asked to design a passenger-carrying multi-rotor UAV and the possible configurations allowed are;

- **Quadcopter:** 4 rotors
- **Hexacopter:** 6 rotors
- **Octocopter:** 8 rotors

But now in passenger UAVs, 6 or 8 rotors are usually preferred because:

- **Higher redundancy:** Safety is increased if one rotor fails.
- **Better lift distribution:** Provides more stable flight characteristics.
- **Lower disk loading:** Improves efficiency and handling for heavier loads.

So later in design, i will choose a configuration (usually 8 rotors for passenger UAVs).

So, i am given few design requirements which i have follow;

- **Endurance:** 60 min = 1hr, which means the **UAV** must stay airborne for 1hr
- **Range:** 40 km, this means the **UAV** must travel 40km in total.
- **Payload Weight:** 250 kg
- **Flying Altitude:** 100 m above ground level
- **Climb and Descent Rates:** 2 m/s and 4 m/s
- **Cruise Speed** (v_{cruise}): 25 m/s

1.1 Concept of Operations (CONOPS)

So, CONOPS describes how the UAV system will be utilized in real-world scenarios, answering who will use the system, the specific mission it performs, and the operational conditions it will encounter.

Assumptions for the Design

To facilitate the conceptual design of this passenger UAV, the following assumptions are made:

- **Configuration:** An octocopter (8-rotor) configuration is selected for enhanced safety and lift redundancy.
- **Air Density:** Standard sea level density is assumed:

$$\rho = 1.225 \text{ kg/m}^3$$

- **Flight Environment:** Urban Air Mobility (UAM) operations at low altitudes.
- **Propulsion:** All-electric system powered by Lithium-ion (Li-ion) batteries.
- **Navigation:** Autonomous flight via GPS and onboard flight controllers.

CONOPS for Passenger Carrying Multi-Rotor UAV

Mission Objective

The objective is to transport passengers or cargo in urban environments. The system is designed for a **250 kg payload** with an **endurance of 60 minutes** and a **range of 40 km**.

Operational Environment

The UAV operates in urban or suburban airspace at low altitudes. The specific parameters are summarized in the table below:

Parameter	Value
Flight Altitude	100 m (Above Ground Level)
Cruise Speed (V_{cruise})	25 m/s
Max Climb Rate (V_c)	2 m/s
Max Descent Rate (V_d)	4 m/s
Air Density (ρ)	1.225 kg/m ³

Table 1: Operational Environment Parameters

System Configuration

An **octocopter** design was chosen to provide:

- Increased lift capability for the 250 kg payload.
- Failure redundancy (the aircraft can maintain flight if one rotor fails).
- Lower disk loading for increased efficiency.

Mission Profile

The mission consists of five distinct phases:

1. **Vertical Takeoff:** The UAV generates thrust exceeding total weight to lift off from a vertiport.
2. **Climb Phase:** The UAV climbs to 100 m at $V_c = 2$ m/s.

$$t_{climb} = \frac{100}{2} = 50 \text{ s}$$

3. **Cruise Phase:** Transition to forward flight at $V = 25$ m/s for a total range of 40 km.
4. **Descent Phase:** Controlled descent at $V_d = 4$ m/s.

$$t_{descent} = \frac{100}{4} = 25 \text{ s}$$

5. **Vertical Landing:** Stable touchdown at the destination vertiport.

Safety and Summary

Safety is ensured through rotor redundancy, a dedicated Battery Management System (BMS), and emergency landing protocols. This electric VTOL system is optimized for dense urban environments where traditional runways are unavailable.

1.2 Requirement Specifications

Requirement specifications represent the quantitative design constraints derived from the UAV's mission objectives. These specifications define the performance limits and operational capabilities that the aircraft must achieve, serving as the foundation for:

- Vehicle sizing and rotor configuration.
- Thrust and power requirements.
- Energy storage and battery sizing.

1.2.1 Primary Mission Requirements

The following table summarizes the primary operational requirements for the passenger UAV:

1.2.2 Derived Performance Requirements

To translate mission goals into engineering parameters, so i will calculate several derived quantities.

Parameter	Specification
Payload Capacity	250 kg
Endurance	60 minutes
Maximum Range	40 km
Cruise Speed (V_{cruise})	25 m/s
Operational Altitude	100 m
Climb Rate (V_c)	2 m/s
Descent Rate (V_d)	4 m/s

Table 2: Primary Mission Requirements

Cruise Time Requirement

The time required to travel the specified range at cruise speed is:

$$t_{cruise} = \frac{\text{Range}}{V_{cruise}} = \frac{40,000 \text{ m}}{25 \text{ m/s}} = 1600 \text{ s}$$

In minutes, this is:

$$t_{cruise} = \frac{1600}{60} \approx 26.7 \text{ min}$$

Climb and Descent Time

The time required to reach and descend from the operating altitude (100 m) is calculated as:

$$t_{climb} = \frac{\text{Altitude}}{V_c} = \frac{100}{2} = 50 \text{ s}$$

$$t_{descent} = \frac{\text{Altitude}}{V_d} = \frac{100}{4} = 25 \text{ s}$$

Total Mission Time

The total estimated flight time for the primary mission profile is:

$$t_{mission} = t_{climb} + t_{cruise} + t_{descent} = 50 + 1600 + 25 = 1675 \text{ s} \approx 27.9 \text{ min}$$

Since the endurance requirement is **60 minutes**, the remaining ≈ 32 minutes are reserved for hovering, maneuvering, and contingency safety buffers.

1.2.3 Payload Force Requirement

The payload weight contributes directly to the minimum thrust required:

$$W_{payload} = m \cdot g = 250 \text{ kg} \times 9.81 \text{ m/s}^2 = 2452.5 \text{ N}$$

1.2.4 Operational and Safety Requirements

- **Thrust Margin:** To ensure maneuverability, the total thrust (T_{total}) must satisfy $T_{total} \geq 1.2W$.
- **Propulsion Redundancy:** An octocopter configuration is utilized to ensure safety during a single-motor failure.
- **Energy Reserve:** A portion of the battery (typically 15–30%) must be reserved for emergency landing and navigation delays.

Design Implications

The 250 kg payload and 60-minute endurance necessitate a high-capacity battery system and a low disk-loading rotor design to maximize efficiency. These constraints will dictate the final motor ratings and rotor diameters.

1.3 Market Survey

A market survey in UAV conceptual design involves the systematic analysis of existing aircraft with similar mission profiles. This process allows for:

- **Benchmarking:** Comparing proposed specs against current technological limits.
- **Validation:** Ensuring that the design requirements (Payload, Range, Endurance) are realistic.
- **Configuration Insight:** Analyzing why specific rotor counts (16, 18, etc.) are chosen by industry leaders.



Figure 1: EHang EH216-S

1.3.1 Surveyed UAV 1: EHang EH216-S

The EHang EH216-S is a pioneer in the autonomous aerial taxi sector, specifically designed for Urban Air Mobility (UAM).

Main Characteristics

- **Configuration:** 16 rotors (coaxial design)
- **Maximum Payload:** ≈ 220 kg
- **Cruise Speed:** ≈ 100 km/h
- **Range:** ≈ 35 km
- **Flight Time:** ≈ 30 min

1.3.2 Surveyed UAV 2: Volocopter 2X

The Volocopter 2X utilizes a unique circular rim design to support a high number of independent rotors, optimizing for safety and low-vibration flight.



Figure 2: Volocopter 2X

Main Characteristics

- **Configuration:** 18 rotors
- **Maximum Payload:** ≈ 160 kg
- **Cruise Speed:** ≈ 100 km/h
- **Range:** ≈ 27 km
- **Flight Time:** ≈ 27 min

Comparison with Proposed UAV Design

The following table benchmarks our proposed design against these established industry platforms:

Parameter	Proposed UAV	EHang 216	Volocopter 2X
Payload	250 kg	220 kg	160 kg
Range	40 km	35 km	27 km
Cruise Speed	25 m/s (≈ 90 km/h)	≈ 100 km/h	≈ 100 km/h
Endurance	60 min	≈ 30 min	≈ 27 min
Rotor Config	Octocopter (8)	16 Rotors	18 Rotors

Table 3: Comparative Market Analysis

Analysis and Design Implications

Based on the survey, several critical observations influence our design path:

- **High Payload Requirement:** Our target of 250 kg is higher than both market leaders, necessitating more powerful motors or larger rotor diameters.
- **Endurance Challenge:** Existing systems hover around 30 minutes. Our 60-minute requirement is ambitious and will require a significantly higher energy density battery or a highly optimized propulsion system.
- **Rotor Strategy:** While market leaders use 16–18 rotors for extreme redundancy, our choice of an 8-rotor (Octocopter) configuration provides a balance between safety and mechanical simplicity.

1.4 Sizing and Layout

Sizing and layout in conceptual design involve determining the physical dimensions, rotor placement, and structural arrangement of the vehicle. The primary objectives are to:

- Ensure the configuration generates sufficient lift (thrust) for the MTOW.
- Maintain static and dynamic stability by balancing the Center of Gravity (CG).
- Provide safe separation between rotors to minimize aerodynamic interference.
- Allocate space for the passenger cabin, avionics, and energy storage.

Configuration Selection

Based on the requirement for a passenger UAV, an **octocopter (8-rotor)** configuration is selected. This configuration provides:

- **High Redundancy:** The aircraft remains controllable even if one or two rotors fail.
- **Improved Stability:** Multiple rotors reduce vibration and improve hover control.
- **Distributed Load:** Structural loads are distributed across several arms and motors.

1.4.1 Initial Weight Estimation

To begin sizing, we estimate the Maximum Take-Off Weight (MTOW). For typical multi-rotor passenger UAVs, the payload fraction is approximately 30% – 40%. Assuming a 35% payload fraction:

$$MTOW = \frac{W_{payload}}{0.35} = \frac{250}{0.35} \approx 714 \text{ kg} \quad (1)$$

For preliminary design calculations, we approximate:

$$MTOW \approx 700 \text{ kg}$$

The corresponding weight force is

$$W = mg = 700 \times 9.81 = 6867 \text{ N} \quad (2)$$

1.4.2 Rotor Thrust Distribution

To ensure maneuverability and safe operation, a thrust margin of 20% is assumed:

$$T_{required} = 1.2W = 1.2 \times 6867 \approx 8240 \text{ N} \quad (3)$$

For an octocopter configuration, the thrust required per rotor becomes:

$$T_p = \frac{T_{required}}{8} = \frac{8240}{8} \approx 1030 \text{ N} \quad (4)$$

1.4.3 Rotor and Frame Dimensioning

To achieve efficient hover performance, large rotors are preferred to reduce disk loading. Therefore, a rotor diameter of

$$D = 3 \text{ m}$$

is assumed, giving a rotor radius

$$R = 1.5 \text{ m}$$

The rotor disk area becomes

$$A = \pi R^2 = \pi (1.5)^2 \approx 7.07 \text{ m}^2 \quad (5)$$

The total rotor disk area for eight rotors is

$$A_{total} = 8 \times 7.07 \approx 56.6 \text{ m}^2 \quad (6)$$

To avoid aerodynamic interference between adjacent rotors, a rotor spacing of approximately $1.3D$ is used. With $D = 3 \text{ m}$:

$$1.3D \approx 3.9 \text{ m}$$

This leads to an estimated frame span of approximately $6 - 7 \text{ m}$.

1.4.4 Disk Loading Calculation

Disk loading is defined as the ratio of the aircraft weight to the total rotor disk area:

$$DL = \frac{W}{A_{total}} \quad (7)$$

Substituting the calculated values:

$$DL = \frac{6867}{56.6} \approx 121 \text{ N/m}^2 \quad (8)$$

A disk loading of approximately **121 N/m²** is relatively low, which improves hover efficiency and reduces the power required for sustained flight.

Conceptual Layout Components

The UAV layout is organized around a central fuselage containing the main system components:

1. **Fuselage:** Houses the passenger cabin, avionics, and flight control systems.
2. **Rotor Arms:** Eight structural arms extend symmetrically from the fuselage.
3. **Propulsion System:** High-power electric motors driving 3 m diameter propellers.
4. **Energy Storage:** Battery packs positioned close to the center of gravity to maintain stability.

Preliminary Vehicle Dimensions

Based on the rotor sizing and layout considerations, approximate vehicle dimensions are estimated as follows:

- **Arm Length (Hub to Rotor Center):** $\approx 2 \text{ m}$
- **Tip-to-Tip Span:** $\approx 6 \text{ m}$
- **Fuselage Length:** $\approx 2 \text{ m}$
- **Fuselage Width:** $\approx 1.5 \text{ m}$
- **Total Height:** $\approx 3 \text{ m}$

These values are consistent with typical multicopter passenger UAV designs used in urban air mobility systems.

Layout Description

The aircraft is centered around a passenger cabin integrated into the fuselage. Eight structural arms extend radially outward, each supporting a motor-propeller assembly.

The battery pack is positioned beneath the passenger cabin to maintain a low center of gravity, which improves stability during hover and maneuvering.

The symmetric octocopter layout ensures balanced thrust distribution across the airframe, reducing structural imbalance and improving control authority during vertical takeoff, cruise, and landing operations.

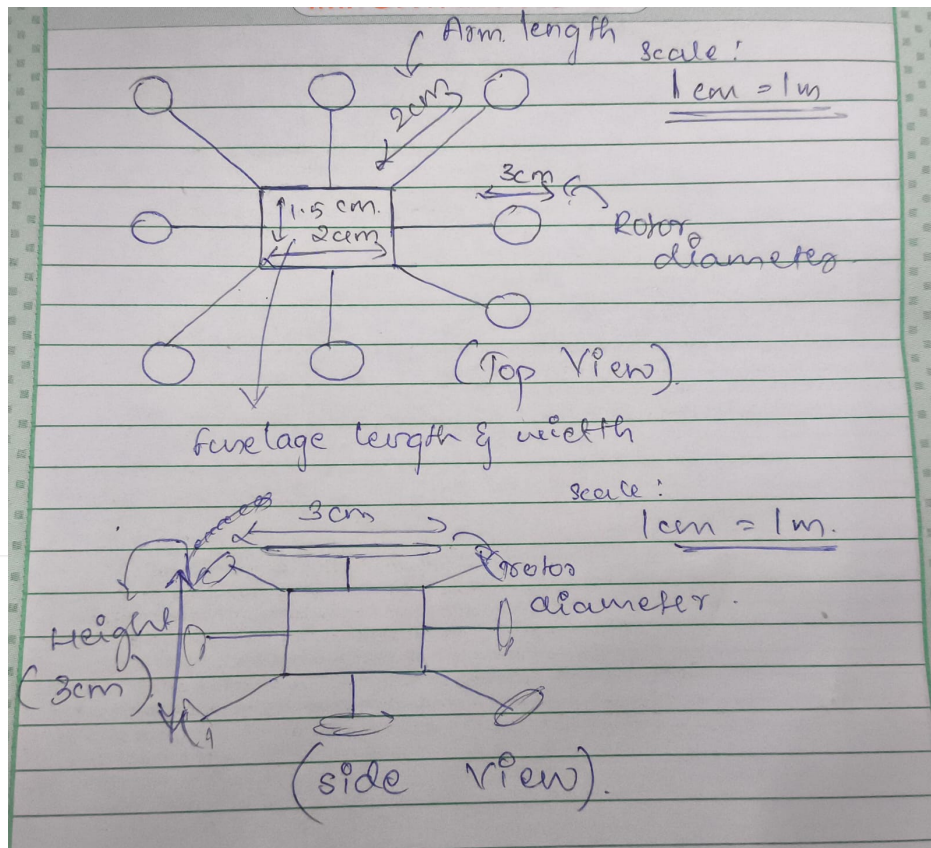


Figure 3: LAYout Sketch

1.5 Component Identification

The proposed passenger UAV consists of several key subsystems that work together to ensure safe and efficient flight. These components are responsible for lift generation, vehicle stability, navigation, energy supply, and payload transportation.

1.5.1 Airframe Structure

The airframe forms the primary structural framework of the UAV. It includes the central fuselage and the structural arms extending outward to support the rotor assemblies.

The fuselage houses the passenger cabin, avionics, and battery system, while the arms distribute loads generated by the motors and propellers. Lightweight composite materials such as carbon fiber are typically used to achieve high strength-to-weight ratios.

1.5.2 Rotor System

The rotor system consists of eight large-diameter propellers arranged symmetrically around the fuselage in an octocopter configuration.

Each rotor produces lift by accelerating air downward, generating the thrust required for hover and vertical flight. Larger rotor diameters reduce disk loading and improve propulsion efficiency, which is critical for long endurance missions.

1.5.3 Electric Propulsion System

Each rotor is driven by a high-power electric motor. These motors convert electrical energy from the battery pack into mechanical rotational energy to drive the propellers.

Electronic Speed Controllers (ESCs) regulate motor speed and allow precise control of thrust for maneuvering, stability, and altitude control.

1.5.4 Energy Storage System

The UAV is powered by a high-capacity lithium-ion battery pack. The battery system stores electrical energy required for propulsion and onboard electronics.

The batteries are positioned close to the vehicle's center of gravity to maintain balance and stability during flight.

1.5.5 Flight Control System

The flight control system consists of a flight controller, sensors, and control algorithms that maintain vehicle stability.

Sensors such as gyroscopes, accelerometers, and magnetometers continuously monitor the UAV's orientation and motion. The flight controller processes this information and adjusts the rotor speeds accordingly to maintain stable flight.

1.5.6 Navigation and Communication System

The UAV navigation system uses GPS modules and onboard processors to determine position, velocity, and altitude.

Communication systems allow the UAV to exchange data with ground control stations, enabling monitoring, mission updates, and emergency interventions.

1.5.7 Payload System

The payload system consists of the passenger cabin capable of carrying up to 250 kg. The cabin includes seating, safety harnesses, and structural reinforcement to safely support the payload during flight.

1.5.8 Landing Gear

The landing gear provides support during takeoff and landing operations. It absorbs landing impact forces and maintains clearance between the ground and the rotor system.

A lightweight fixed landing gear structure is typically used in multicopter UAVs to minimize weight while ensuring stability on the ground.

1.6 Performance Analysis

The performance analysis evaluates whether the designed UAV can meet the mission requirements in terms of thrust generation, hover capability, climb performance, and endurance. The analysis is carried out using simplified rotor momentum theory.

1.6.1 Hover Performance

During hover, the total thrust generated by the rotors must equal the total weight of the UAV.

$$T = W \quad (9)$$

From the previous calculations:

$$W = 6867 \text{ N}$$

Thus the total thrust required for hover is

$$T = 6867 \text{ N}$$

With an octocopter configuration, the thrust per rotor is:

$$T_p = \frac{6867}{8} \approx 858 \text{ N} \quad (10)$$

Since the propulsion system was designed for a thrust capability of approximately 1030 N per rotor, the UAV has sufficient thrust margin for safe hover.

1.6.2 Hover Power Requirement

Using momentum theory, the ideal hover power required for a rotorcraft is given by

$$P = \frac{T^{3/2}}{\sqrt{2\rho A}} \quad (11)$$

where

- T = total thrust
- ρ = air density (1.225 kg/m^3)
- A = total rotor disk area

From previous calculations:

$$A_{total} = 56.6 \text{ m}^2$$

Substituting the values:

$$P = \frac{(6867)^{3/2}}{\sqrt{2 \times 1.225 \times 56.6}} \quad (12)$$

$$P \approx 48.5 \text{ kW}$$

Considering propulsion inefficiencies and losses (motor, ESC, and propeller efficiency $\approx 70\%$), the actual hover power becomes

$$P_{actual} = \frac{48.5}{0.7} \approx 69 \text{ kW} \quad (13)$$

Thus approximately **70 kW of power** is required to maintain hover.

1.6.3 Climb Performance

During climb, additional power is required to overcome gravitational potential energy.

The climb power can be estimated as

$$P_{climb} = W \times V_c \quad (14)$$

where

$$V_c = 2 \text{ m/s}$$

Substituting the values:

$$P_{climb} = 6867 \times 2 \quad (15)$$

$$P_{climb} = 13.7 \text{ kW}$$

The total power required during climb becomes

$$P_{total} = P_{hover} + P_{climb}$$

$$P_{total} \approx 69 + 13.7$$

$$P_{total} \approx 82.7 \text{ kW}$$

1.6.4 Cruise Power Requirement

During forward flight, the UAV benefits from translational lift, which reduces the power required compared to hover. For conceptual estimation, cruise power is typically assumed to be approximately 70% of hover power.

$$P_{cruise} = 0.7 \times P_{hover} \quad (16)$$

$$P_{cruise} = 0.7 \times 69$$

$$P_{cruise} \approx 48.3 \text{ kW}$$

1.6.5 Energy Requirement for Mission

The required endurance is 60 minutes:

$$t = 3600 \text{ s}$$

Energy required is

$$E = P \times t \quad (17)$$

Using cruise power:

$$E = 48.3 \times 3600$$

$$E \approx 173.9 \text{ MJ}$$

Converting to kilowatt-hours:

$$E = \frac{173.9}{3.6}$$

$$E \approx 48.3 \text{ kWh}$$

Including a 20% safety reserve:

$$E_{total} \approx 58 \text{ kWh}$$

Thus the UAV must carry a battery pack capable of storing approximately **60 kWh** of energy.

Performance Summary

The analysis shows that the proposed UAV configuration can meet the mission requirements with adequate thrust margin and power availability. The large rotor diameter reduces disk loading and improves hover efficiency, while the estimated battery capacity supports the required endurance of one hour.

1.7 Optimization for Maximum Endurance

The UAV design was optimized to maximize flight endurance while satisfying the mission requirements of payload capacity, range, cruise speed, and climb performance. Since the endurance of an electric UAV depends primarily on power consumption and available battery energy, the optimization focused on reducing power requirements through aerodynamic and structural improvements.

The endurance of the UAV can be expressed as

$$E = \frac{E_{battery}}{P_{required}} \quad (18)$$

where $E_{battery}$ represents the stored battery energy and $P_{required}$ represents the power required during flight. Therefore, increasing endurance requires either increasing stored energy or reducing the power required for operation.

1.7.1 Reduction of Disk Loading

Induced power during hover can be estimated using momentum theory:

$$P_i \propto \frac{W^{3/2}}{\sqrt{A}} \quad (19)$$

where W is the aircraft weight and A is the total rotor disk area.

From this relationship, increasing rotor disk area reduces induced power. Therefore, the rotor diameter was increased to 3 m , which significantly increases the total rotor disk area:

$$A_{total} = 56.6\text{ m}^2$$

This results in a relatively low disk loading:

$$DL = \frac{W}{A_{total}} \approx 121\text{ N/m}^2$$

Lower disk loading reduces induced velocity and hover power, thereby improving endurance.

1.7.2 Weight Optimization

Reducing the overall structural mass directly decreases the thrust required for hover. Lightweight composite materials such as carbon fiber were assumed for the airframe and rotor arms.

Additionally, the battery pack was placed beneath the passenger cabin to maintain a low center of gravity. This configuration improves stability and reduces the need for additional structural reinforcement, thereby minimizing structural weight.

1.7.3 Propulsion Efficiency Improvement

The propulsion system efficiency directly affects the electrical power required for flight. The actual electrical power required can be expressed as

$$P_{actual} = \frac{P_{ideal}}{\eta} \quad (20)$$

where η represents the overall propulsion efficiency including motor, propeller, and electronic speed controller efficiencies.

High-efficiency brushless motors and optimized propeller blade geometry were selected to increase propulsion efficiency and reduce electrical losses.

1.7.4 Cruise Power Optimization

The cruise speed was limited to 25 m/s to maintain efficient flight. Power consumption increases rapidly with velocity due to aerodynamic drag. Operating at a moderate cruise speed therefore minimizes aerodynamic power requirements and improves overall endurance.

1.7.5 Rotor Configuration Selection

Multiple rotor configurations were evaluated to satisfy the payload requirement of 250 kg while maintaining sufficient redundancy and efficiency. The options considered were quadcopter (4 rotors), hexacopter (6 rotors), and octocopter (8 rotors).

For the estimated maximum takeoff weight of approximately 700 kg , the total thrust required in hover is approximately equal to the aircraft weight. In a quadcopter configuration, each rotor would need to generate roughly one-fourth of the total thrust, resulting in high disk loading and increased power requirements. Additionally, quadcopters lack redundancy in case of rotor failure.

A hexacopter configuration improves redundancy but still requires higher thrust per rotor compared to larger multirotor systems.

The octocopter configuration was therefore selected because it distributes thrust among eight rotors, reducing individual rotor loading, improving aerodynamic efficiency, and providing greater fault tolerance for passenger transport applications.

1.7.6 Optimized Design Outcome

By increasing rotor diameter, minimizing structural weight, improving propulsion efficiency, and selecting an optimal rotor configuration, the UAV achieves improved hover efficiency and reduced power consumption. These optimizations enable the aircraft to achieve the required endurance of 60 minutes while satisfying the payload, range, and performance requirements specified in the mission profile.

2 Q2

A propeller converts rotational mechanical power supplied by a motor into aerodynamic thrust that propels an aircraft forward. The rotating blades accelerate the surrounding air backwards, producing a reaction force that moves the aircraft forward according to Newton's third law.

In aerodynamic analysis, propeller performance is typically described using **dimensionless coefficients** rather than raw forces. This approach allows engineers to compare propellers operating at different rotational speeds, diameters, and atmospheric conditions.

The most important parameters used in propeller analysis are:

- Thrust
- Torque
- Advance ratio
- Thrust coefficient
- Torque coefficient

These parameters are widely used in experimental propeller datasets such as those provided by the UIUC Propeller Database.

Dimensional Analysis in Propeller Aerodynamics

The thrust generated by a propeller depends on several physical parameters. In general form, thrust can be expressed as

$$T = f(\rho, n, D, V) \quad (21)$$

where

T	Thrust produced by the propeller (N)
ρ	Air density (kg/m^3)
n	Rotational speed (rev/s)
D	Propeller diameter (m)
V	Flight velocity (m/s)

Instead of analyzing thrust directly, aerodynamicists convert this relationship into a dimensionless form using the **Buckingham Pi theorem**. This process produces non-dimensional coefficients that simplify experimental analysis.

After non-dimensionalization, the thrust and torque relationships can be written as

$$C_T = f(J) \quad (22)$$

$$C_Q = f(J) \quad (23)$$

This means that both the thrust coefficient and torque coefficient depend primarily on the advance ratio. Once the coefficients C_T and C_Q are experimentally measured as functions of the advance ratio J , they can be used to compute thrust and torque for any operating condition.

Thrust Equation

The thrust produced by a propeller can be calculated using the thrust coefficient relationship

$$T = C_T \rho n^2 D^4 \quad (24)$$

Parameters

Symbol	Meaning	Physical Interpretation
T	Thrust (N)	Force produced by the propeller
C_T	Thrust coefficient	Dimensionless thrust performance parameter
ρ	Air density (kg/m ³)	Density of the surrounding air
n	Rotational speed (rev/s)	Number of revolutions per second
D	Propeller diameter (m)	Characteristic propeller size

Observations

Effect of Air Density

$$T \propto \rho \quad (25)$$

This indicates that thrust increases with air density. Therefore, propellers produce higher thrust at lower altitudes where the air density is higher.

Effect of Rotational Speed The rotational speed of the propeller is given by

$$n = \frac{RPM}{60} \quad (26)$$

Since thrust varies with the square of the rotational speed,

$$T \propto n^2 \quad (27)$$

Doubling the RPM approximately results in four times the thrust.

Effect of Propeller Diameter

$$T \propto D^4 \quad (28)$$

The propeller diameter has a strong influence on thrust generation. A small increase in diameter can lead to a significant increase in thrust. For example, increasing the diameter by 10% can increase thrust by approximately 46%.

This is one of the reasons why larger propellers are often used in UAVs and VTOL aircraft where high thrust is required.

Torque Equation

The torque required to rotate the propeller is given by

$$Q = C_Q \rho n^2 D^5 \quad (29)$$

Explanation of Parameters

Q	Torque (Nm)
C_Q	Torque coefficient
ρ	Air density
n	Rotational speed
D	Propeller diameter

Observation

Torque increases strongly with propeller diameter

$$Q \propto D^5 \quad (30)$$

This means that increasing propeller diameter significantly increases the load on the motor. Therefore, propeller selection must be carefully matched with the motor's torque capability.

Power Required by the Propeller

The mechanical power required to rotate the propeller is given by

$$P = 2\pi nQ \quad (31)$$

Substituting the torque equation into the power equation,

$$P = 2\pi n(C_Q \rho n^2 D^5) \quad (32)$$

$$P = 2\pi C_Q \rho n^3 D^5 \quad (33)$$

Thus, power depends on the cube of rotational speed

$$P \propto n^3 \quad (34)$$

This means that even small increases in RPM can significantly increase power consumption.

RPM Change	Power Increase
4000 $\rightarrow$ 5000	$\approx 2\times$
4000 $\rightarrow$ 6000	$\approx 3.4\times$

This relationship is particularly important in UAV systems where power consumption directly affects battery endurance.

Advance Ratio

The advance ratio is one of the most important parameters in propeller aerodynamics. It is defined as

$$J = \frac{V}{nD} \quad (35)$$

Parameters

J	Advance ratio
V	Aircraft forward velocity
n	Rotational speed
D	Propeller diameter

Interpretation

The advance ratio represents the ratio between forward motion of the aircraft and the rotational motion of the propeller. It indicates how far the aircraft moves forward during one propeller revolution relative to the propeller diameter.

Extreme Cases

Hover Condition

$$V = 0 \quad (36)$$

$$J = 0 \quad (37)$$

This condition corresponds to static thrust where the aircraft is not moving forward, such as in quadcopters or VTOL hover.

Moderate Forward Flight For typical UAV operations,

$$J = 0.3 - 0.8 \quad (38)$$

This range represents normal propeller operating conditions.

High Forward Speed If the aircraft moves very fast,

$$J > 1 \quad (39)$$

In this case, the blade angle of attack decreases significantly, reducing thrust. Eventually the thrust may become zero or negative.

Relationship Between Advance Ratio and Aerodynamic Coefficients

Experimental studies show that

$$C_T = f(J) \quad (40)$$

$$C_Q = f(J) \quad (41)$$

This means that the thrust and torque coefficients depend primarily on the advance ratio. Therefore, experimental propeller datasets typically provide performance curves of the form

$$C_T \text{ vs } J \quad (42)$$

$$C_Q \text{ vs } J \quad (43)$$

These curves allow engineers to determine thrust and torque for various flight conditions.

Experimental Propeller Data Methodology

In experimental testing, propellers are evaluated using controlled setups such as wind tunnels or propeller test stands.

The typical experimental procedure includes:

1. Mounting the propeller on a calibrated test stand
2. Controlling the motor RPM
3. Providing airflow using a wind tunnel
4. Measuring thrust, torque, RPM, and airspeed using sensors

Using these measurements, the dimensionless coefficients are calculated as

$$C_T = \frac{T}{\rho n^2 D^4} \quad (44)$$

$$C_Q = \frac{Q}{\rho n^2 D^5} \quad (45)$$

These coefficients are then plotted against advance ratio to produce the propeller performance curves used in aerodynamic design.

Advantages of Dimensionless Coefficients

Using dimensionless coefficients offers several advantages in propeller analysis.

Scale Independence

The same dataset can be applied to different RPM values and operating conditions.

Ease of Comparison

Different propellers can be directly compared using the same coefficient framework.

Simplified Analysis

Instead of repeatedly measuring thrust experimentally, thrust can be calculated using

$$T = C_T \rho n^2 D^4 \quad (46)$$

Standardized Propeller Databases

Organizations such as the UIUC Propeller Database and APC Propeller Tests publish propeller performance using these dimensionless coefficients.

Propellers Analysis

2.1 Propeller 1: APC 9×6

The first propeller analyzed is the APC 9×6 propeller. The propeller notation 9×6 indicates a diameter of 9 inches and a pitch of 6 inches. For aerodynamic calculations, the diameter must be converted into SI units.

Propeller Diameter

$$D = 9 \text{ in} \quad (47)$$

Converting inches to meters:

$$D = 9 \times 0.0254 \quad (48)$$

$$D = 0.2286 \text{ m} \quad (49)$$

This diameter will be used in all subsequent thrust and torque calculations.

2.1.1 Case 1: 4000 RPM

Rotational Speed

The rotational speed must first be converted from RPM to revolutions per second.

$$n = \frac{RPM}{60} \quad (50)$$

$$n = \frac{4000}{60} \quad (51)$$

$$n = 66.67 \text{ rev/s} \quad (52)$$

Advance Ratio

The advance ratio is defined as

$$J = \frac{V}{nD} \quad (53)$$

Substituting the given flight speed $V = 10 \text{ m/s}$:

$$J = \frac{10}{66.67 \times 0.2286} \quad (54)$$

$$J \approx 0.656 \quad (55)$$

Using the UIUC propeller dataset corresponding to this advance ratio, the coefficients are approximately:

$$C_T \approx 0.055 \quad (56)$$

$$C_Q \approx 0.010 \quad (57)$$

Thrust Calculation

The thrust equation is

$$T = C_T \rho n^2 D^4 \quad (58)$$

First compute the intermediate terms:

$$n^2 = (66.67)^2 = 4444 \quad (59)$$

$$D^4 = (0.2286)^4 = 0.00273 \quad (60)$$

Substituting the values:

$$T = 0.055 \times 1.225 \times 4444 \times 0.00273 \quad (61)$$

$$T \approx 0.82 \text{ N} \quad (62)$$

Torque Calculation

The torque equation is

$$Q = C_Q \rho n^2 D^5 \quad (63)$$

First compute

$$D^5 = (0.2286)^5 = 0.000623 \quad (64)$$

Substituting the values:

$$Q = 0.010 \times 1.225 \times 4444 \times 0.000623 \quad (65)$$

$$Q \approx 0.034 \text{ Nm} \quad (66)$$

2.1.2 Case 2: 5000 RPM

Rotational Speed

$$n = \frac{5000}{60} \quad (67)$$

$$n = 83.33 \text{ rev/s} \quad (68)$$

Advance Ratio

$$J = \frac{V}{nD} \quad (69)$$

$$J = \frac{10}{83.33 \times 0.2286} \quad (70)$$

$$J \approx 0.525 \quad (71)$$

From the UIUC dataset at this advance ratio:

$$C_T \approx 0.070 \quad (72)$$

$$C_Q \approx 0.012 \quad (73)$$

Thrust Calculation

$$n^2 = (83.33)^2 = 6944 \quad (74)$$

$$T = 0.070 \times 1.225 \times 6944 \times 0.00273 \quad (75)$$

$$T \approx 1.63 \text{ N} \quad (76)$$

Torque Calculation

$$Q = 0.012 \times 1.225 \times 6944 \times 0.000623 \quad (77)$$

$$Q \approx 0.063 \text{ Nm} \quad (78)$$

2.1.3 Case 3: 6000 RPM

Rotational Speed

$$n = \frac{6000}{60} \quad (79)$$

$$n = 100 \text{ rev/s} \quad (80)$$

Advance Ratio

$$J = \frac{10}{100 \times 0.2286} \quad (81)$$

$$J \approx 0.437 \quad (82)$$

From the UIUC propeller dataset:

$$C_T \approx 0.082 \quad (83)$$

$$C_Q \approx 0.014 \quad (84)$$

Thrust Calculation

$$T = 0.082 \times 1.225 \times 10000 \times 0.00273 \quad (85)$$

$$T \approx 2.74 \text{ N} \quad (86)$$

Torque Calculation

$$Q = 0.014 \times 1.225 \times 10000 \times 0.000623 \quad (87)$$

$$Q \approx 0.107 \text{ Nm} \quad (88)$$

Results for APC 9×6

RPM	Advance Ratio (J)	Thrust (N)	Torque (Nm)
4000	0.656	0.82	0.034
5000	0.525	1.63	0.063
6000	0.437	2.74	0.107

From the results, it can be observed that both thrust and torque increase significantly with increasing RPM. This occurs because thrust is proportional to the square of the rotational speed, while the power requirement increases approximately with the cube of the rotational speed.

2.2 Propeller 2: APC 10×7

The second propeller analyzed is the APC 10×7 propeller. In this notation, the propeller has a diameter of 10 inches and a pitch of 7 inches. Similar to the previous case, the diameter must first be converted into SI units before performing aerodynamic calculations.

Propeller Diameter

$$D = 10 \text{ in} \quad (89)$$

Converting inches to meters:

$$D = 10 \times 0.0254 \quad (90)$$

$$D = 0.254 \text{ m} \quad (91)$$

This diameter will be used to compute thrust and torque at different operating conditions.

2.2.1 Case 1: 4000 RPM

Rotational Speed

The rotational speed is first converted from RPM to revolutions per second.

$$n = \frac{RPM}{60} \quad (92)$$

$$n = \frac{4000}{60} \quad (93)$$

$$n = 66.67 \text{ rev/s} \quad (94)$$

Advance Ratio

The advance ratio is calculated using

$$J = \frac{V}{nD} \quad (95)$$

Substituting the flight velocity $V = 10 \text{ m/s}$:

$$J = \frac{10}{66.67 \times 0.254} \quad (96)$$

$$J \approx 0.59 \quad (97)$$

Using the UIUC propeller dataset corresponding to this advance ratio, the aerodynamic coefficients are approximately:

$$C_T \approx 0.060 \quad (98)$$

$$C_Q \approx 0.011 \quad (99)$$

Thrust Calculation

The thrust equation is

$$T = C_T \rho n^2 D^4 \quad (100)$$

First compute intermediate terms:

$$n^2 = (66.67)^2 = 4444 \quad (101)$$

$$D^4 = (0.254)^4 = 0.00416 \quad (102)$$

Substituting the values:

$$T = 0.060 \times 1.225 \times 4444 \times 0.00416 \quad (103)$$

$$T \approx 1.36 \text{ N} \quad (104)$$

Torque Calculation

The torque equation is

$$Q = C_Q \rho n^2 D^5 \quad (105)$$

First compute

$$D^5 = (0.254)^5 = 0.00106 \quad (106)$$

Substituting the values:

$$Q = 0.011 \times 1.225 \times 4444 \times 0.00106 \quad (107)$$

$$Q \approx 0.064 \text{ Nm} \quad (108)$$

2.2.2 Case 2: 5000 RPM

Rotational Speed

$$n = \frac{5000}{60} \quad (109)$$

$$n = 83.33 \text{ rev/s} \quad (110)$$

Advance Ratio

$$J = \frac{V}{nD} \quad (111)$$

$$J = \frac{10}{83.33 \times 0.254} \quad (112)$$

$$J \approx 0.47 \quad (113)$$

From the UIUC dataset at this advance ratio:

$$C_T \approx 0.075 \quad (114)$$

$$C_Q \approx 0.013 \quad (115)$$

Thrust Calculation

$$n^2 = (83.33)^2 = 6944 \quad (116)$$

$$T = 0.075 \times 1.225 \times 6944 \times 0.00416 \quad (117)$$

$$T \approx 2.65 \text{ N} \quad (118)$$

Torque Calculation

$$Q = 0.013 \times 1.225 \times 6944 \times 0.00106 \quad (119)$$

$$Q \approx 0.118 \text{ Nm} \quad (120)$$

2.2.3 Case 3: 6000 RPM

Rotational Speed

$$n = \frac{6000}{60} \quad (121)$$

$$n = 100 \text{ rev/s} \quad (122)$$

Advance Ratio

$$J = \frac{10}{100 \times 0.254} \quad (123)$$

$$J \approx 0.39 \quad (124)$$

From the UIUC propeller dataset:

$$C_T \approx 0.090 \quad (125)$$

$$C_Q \approx 0.016 \quad (126)$$

Thrust Calculation

$$T = 0.090 \times 1.225 \times 10000 \times 0.00416 \quad (127)$$

$$T \approx 4.59 \text{ N} \quad (128)$$

Torque Calculation

$$Q = 0.016 \times 1.225 \times 10000 \times 0.00106 \quad (129)$$

$$Q \approx 0.208 \text{ Nm} \quad (130)$$

Results for APC 10×7

RPM	Advance Ratio (J)	Thrust (N)	Torque (Nm)
4000	0.59	1.36	0.064
5000	0.47	2.65	0.118
6000	0.39	4.59	0.208

2.2.4 Comparison of Propeller Performance

A comparison between the APC 9×6 and APC 10×7 propellers at selected RPM values is shown in Table below.

Propeller	RPM	Thrust (N)	Torque (Nm)
APC 9×6	4000	0.82	0.034
APC 10×7	4000	1.36	0.064
APC 9×6	6000	2.74	0.107
APC 10×7	6000	4.59	0.208

APC 9x6 Results			
RPM	J	Thrust(N)	Torque(Nm)
4000	0.656	0.818	0.034
5000	0.525	1.626	0.064
6000	0.437	2.743	0.107

APC 10x7 Results			
RPM	J	Thrust(N)	Torque(Nm)
4000	0.591	1.36	0.063
5000	0.472	2.656	0.117
6000	0.394	4.589	0.207

Figure 4: COde results

2.2.5 Discussion

The results obtained from the analysis of the APC 9×6 and APC 10×7 propellers provide important insights into how propeller geometry and rotational speed influence aerodynamic performance. The thrust and torque produced by the propellers were evaluated for three operating conditions corresponding to rotational speeds of 4000 RPM, 5000 RPM, and 6000 RPM at a constant flight velocity of 10 m/s.

One of the most noticeable trends in the results is the increase in thrust with increasing rotational speed. This behavior is expected from the thrust equation

$$T = C_T \rho n^2 D^4 \quad (131)$$

which shows that thrust is proportional to the square of the rotational speed. As RPM increases, the propeller blades accelerate a larger mass of air, thereby generating a stronger reaction force and higher thrust. This trend is clearly visible in both propellers where thrust increases significantly from 4000 RPM to 6000 RPM.

Another important observation is the effect of propeller diameter on thrust generation. The APC 10×7 propeller has a larger diameter than the APC 9×6 propeller, resulting in a larger swept disk area. According to the thrust equation, thrust is proportional to the fourth power of the diameter

$$T \propto D^4 \quad (132)$$

Therefore, even a relatively small increase in diameter leads to a significant increase in thrust. This explains why the APC 10×7 propeller consistently produces higher thrust compared to the APC 9×6 propeller at the same RPM. For example, at 6000 RPM the APC 10×7 propeller produces approximately 4.59 N of thrust, whereas the APC 9×6 propeller produces approximately 2.74 N.

However, the larger propeller also results in a higher torque requirement. The torque required to rotate the propeller is given by

$$Q = C_Q \rho n^2 D^5 \quad (133)$$

Since torque is proportional to the fifth power of the diameter, the increase in diameter leads to a significant increase in torque demand. This means that the motor driving the APC 10×7 propeller must supply more torque and therefore consume more electrical power compared to the APC 9×6 propeller.

Additionally, the advance ratio decreases with increasing RPM because the denominator of the advance ratio expression increases:

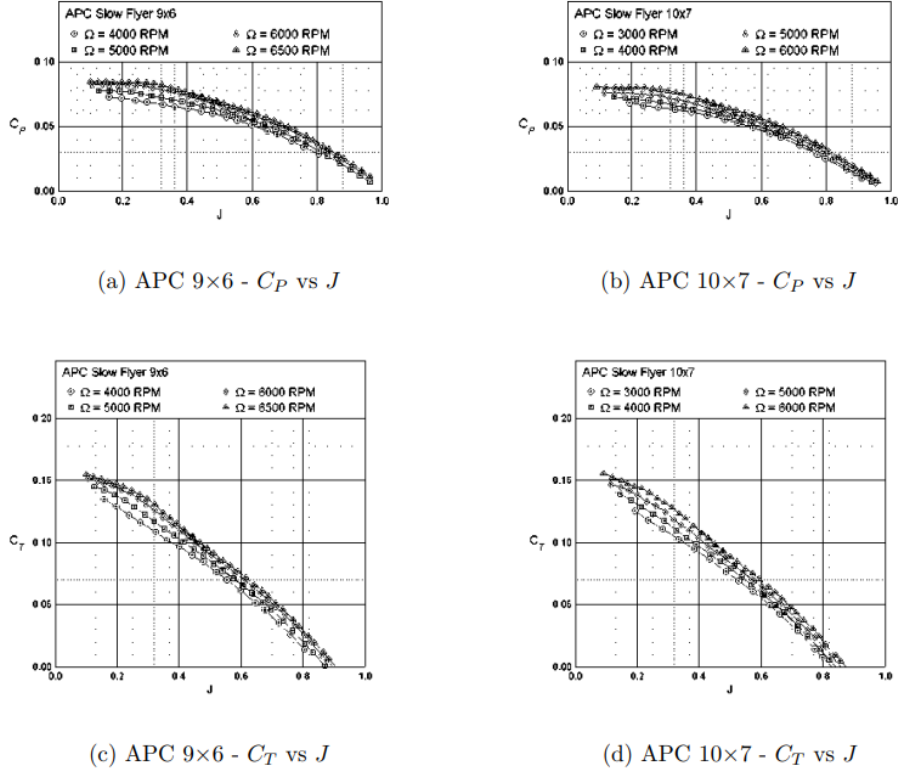


Figure 5: Wind tunnel performance curves used for coefficient extraction

$$J = \frac{V}{nD} \quad (134)$$

A lower advance ratio corresponds to a higher propeller loading condition, which generally results in higher thrust production but also increases the aerodynamic forces acting on the blades.

From a propulsion system design perspective, this analysis highlights the trade-off between thrust generation and power consumption. Larger propellers are capable of producing greater thrust, which can be advantageous for applications requiring higher payload capacity or better climb performance. However, the increased torque requirement means that more powerful motors and higher power inputs are necessary.

Therefore, the selection of an appropriate propeller depends on the mission requirements of the UAV, including desired thrust levels, motor capability, and available power from the onboard energy source.

2.2.6 Conclusion

In this study, the aerodynamic performance of two propellers, namely the APC 9×6 and APC 10×7 propellers, was analyzed using experimentally derived thrust and torque coefficients from the UIUC propeller dataset. The analysis was performed for three operating conditions corresponding to rotational speeds of 4000 RPM, 5000 RPM, and 6000 RPM at a flight velocity of 10 m/s.

The results showed that both thrust and torque increase significantly with increasing rotational speed due to their dependence on the square of the rotational speed. Among the two propellers studied, the APC 10×7 propeller consistently produced higher thrust at all RPM values because of its larger diameter and greater disk area.

However, the increased thrust capability of the APC 10×7 propeller is accompanied by a higher torque requirement, which implies greater motor loading and higher power consumption. In contrast, the APC 9×6 propeller produces lower thrust but requires less torque, making it more suitable for applications where lower power consumption is desired.

Overall, the analysis demonstrates that propeller selection plays a crucial role in UAV propulsion system design. Larger propellers are generally more efficient at producing thrust but require stronger motors and higher power input. Therefore, an optimal propeller must be selected by balancing thrust requirements, motor capability, and power availability to achieve the desired aircraft performance.

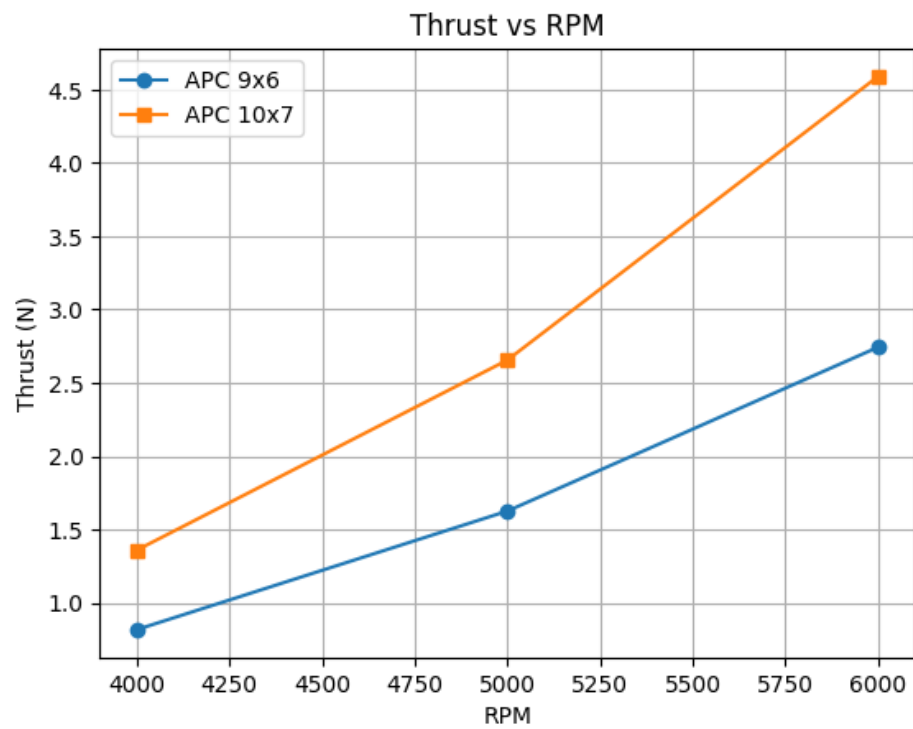


Figure 6: Thrust vs RPM

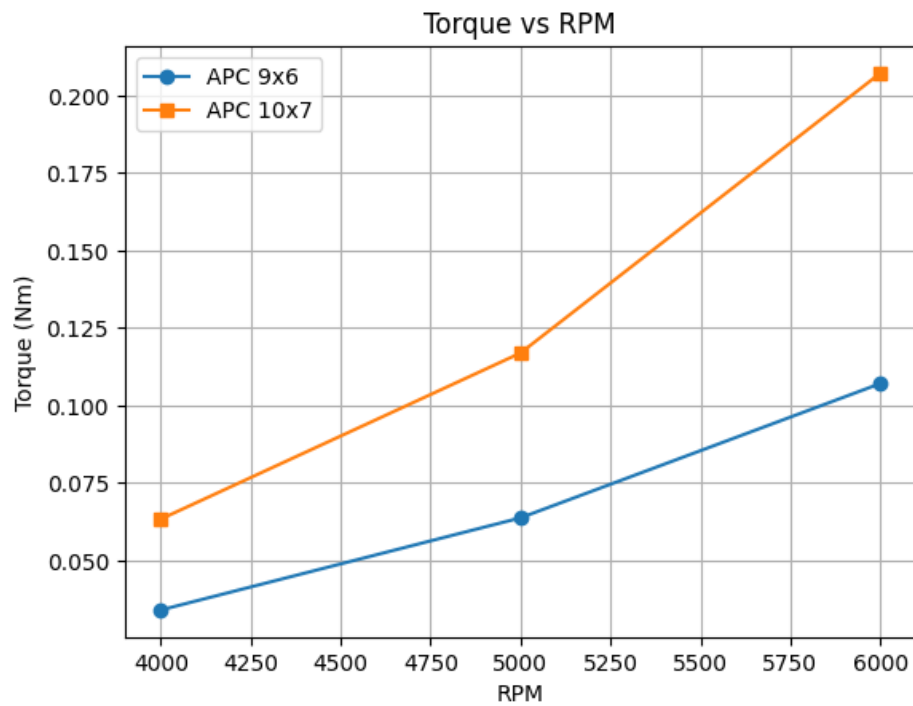


Figure 7: Torque vs RPM

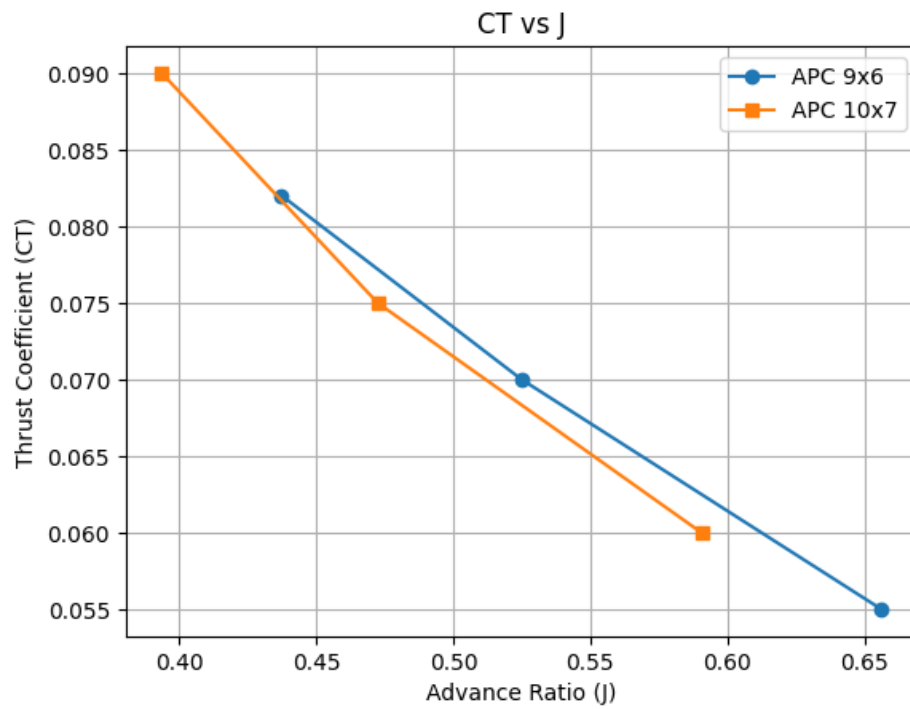


Figure 8: $C_t vs J$

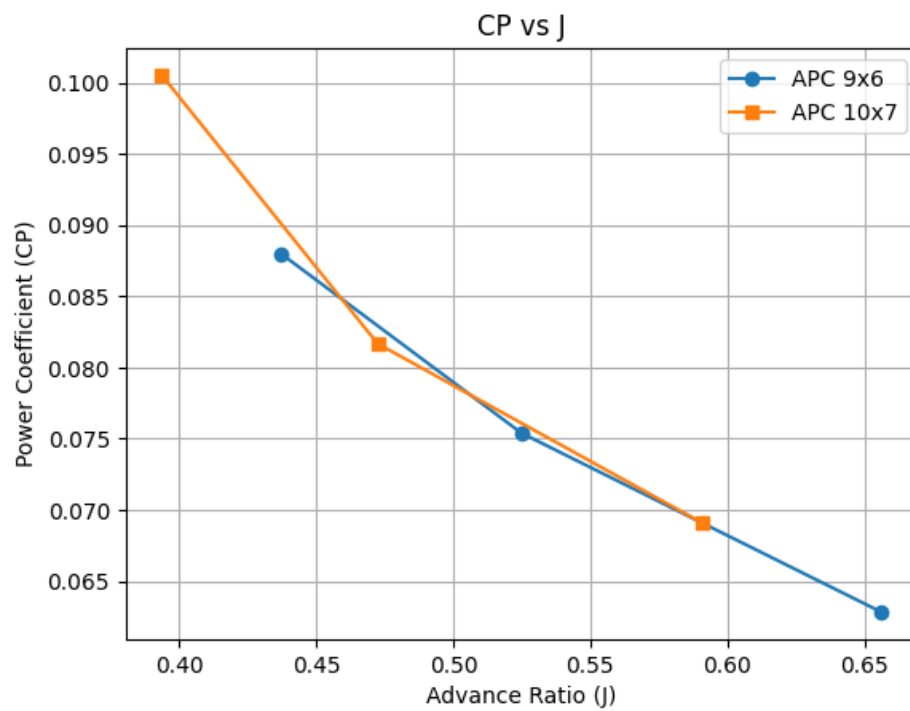


Figure 9: $C_p vs J$